

Low Dose and High Contrast Biomedical Imaging Using Self-Supervised Deep Learning

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Abstract

Self-supervised deep learning has emerged as a powerful method for image enhancement when *a priori* ground-truth references are not available. Stemming from Noise2Noise¹, it was shown that a convolutional neural network (CNN) can be trained from a noisy input and target pair of the same scene to produce clean images, given that the noise distributions are independent and image features have the same mean grey value (Figure 1 A). However, while this method has shown great promise for denoising, the majority of literature since has been largely focused on noise reduction. Addressing other prevalent artefacts in biomedical imaging, such as high-contrast visualization of soft tissues and low-dose imaging, depends on image quality beyond noise.

We have proposed that the underlying principle, that a CNN trained on independently corrupted pairs of images depicting the same scene can be used to recover shared structure. This generalizes to self-supervised deep-learning as an artefact reduction tool for ring artefacts, sparse-view streaks, and contrast enhancement. This is supported by our previous works, EVEPR² departed from the mean grey value convention by pairing edge-enhanced contrast inputs with phase-retrieved targets, training a CNN to recover phase contrast and reduce over-smoothing (Figure 1 B). Sparse2Noise³ showed that input and target pairs with different sparse-view streak patterns and different ring artefact levels can reduce and suppress those respective artefacts (Figure 1 C and D).

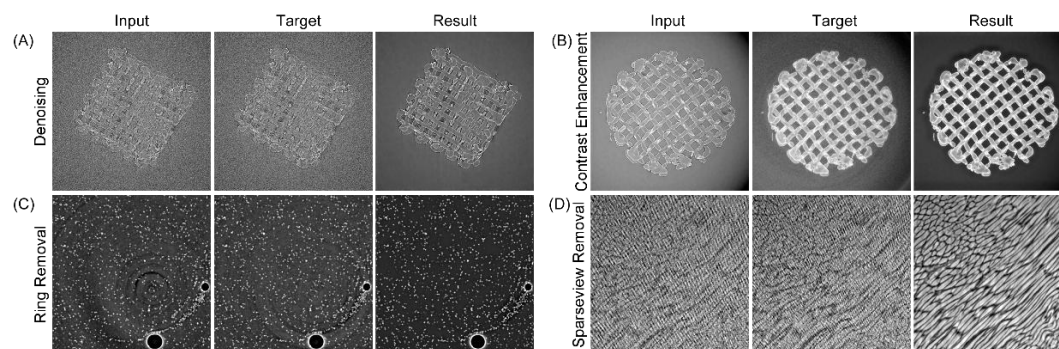


Figure 1: Representative input–target pairs and CNN results for (A) denoising, (B) low density material contrast enhancement, (C) ring artefact removal, and (D) sparse view streak artefact removal

Each targeted artefact carries distinct imaging consequences. Noise reduction improves signal-to-noise ratio (SNR) and contrast-to-noise ratio (CNR). Ring artefact suppression improves material discriminability. Enhanced phase contrast is especially impactful for low-density materials such as soft tissues. Sparse-view artefact reduction enables high-quality reconstruction from fewer projections, directly supporting fast and low-dose imaging protocols. Together, these applications make self-supervised deep learning a broadly applicable tool for synchrotron-based biomedical imaging because training only requires careful preprocessed input and target pairs derived from the same acquisition, without *a priori* reference.

[1] Lehtinen J et al. Noise2Noise: Learning Image Restoration without Clean Data. <http://arxiv.org/abs/1803.04189>, [2] Duan X et al. Sparse2Noise: Low-dose synchrotron X-ray tomography without high-quality reference data. *Comput Biol Med.* 2023;165. doi:10.1016/j.combiomed.2023.107473, [3] Ding XF et al. Development of a deep learning method for phase retrieval image enhancement in phase contrast microcomputed tomography. *J Microsc.* doi:10.1111/jmi.13419